ANSWER KEY & MARKING SCHEME — MATHEMATICS · CLASS 7 · SET 01

Max. Marks: 60 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Multiple Choice Questions (12 × 1 = 12) — 1 questions · 12 marks

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) One billion is equal to how many crore?

(a) 100 crore (b) 10 crore (c) 1000 crore (d) 1 crore

> **Ans: (a)** 1 billion = 10^9 and 1 crore = 10^7, so 1 billion = 100 crore.

(ii) The number of seconds in one day is:

(a) 14,400 (b) 3,600 (c) 8,640 (d) 86,400

> **Ans: (d)** 24 × 60 × 60 = 86,400.

(iii) The value of 36 ÷ 6 × 3 + 2 is:

(a) 4 (b) 30 (c) 20 (d) 8

> **Ans: (c)** Left to right: 36 ÷ 6 = 6; 6 × 3 = 18; 18 + 2 = 20.

(iv) In the decimal number 4.058, the digit 5 is in the:

(a) hundredths place (b) tenths place
(c) thousandths place (d) tens place

> **Ans: (a)** 4 . 0 5 8: tenths = 0, hundredths = 5, thousandths = 8.

(v) Which of these decimals is the greatest?

(a) 0.49 (b) 0.499 (c) 0.05 (d) 0.5

> **Ans: (d)** 0.5 = 0.500 > 0.499 > 0.490 > 0.050.

(vi) 'Five more than three times a number n' is written as:

(a) 5n + 3 (b) 3(n + 5) (c) 3n + 5 (d) 3n − 5

> **Ans: (c)** Three times n is 3n; five more gives 3n + 5.

(vii) The value of $2a^2 - 3b$ when a = 3 and b = 2 is:

(a) 6 (b) 0 (c) 12 (d) 24

> **Ans: (c)** 2 × 9 − 3 × 2 = 18 − 6 = 12.

(viii) One angle of a linear pair is 65°. The other angle is:

(a) 25° (b) 125° (c) 295° (d) 115°

> **Ans: (d)** Angles of a linear pair add up to 180°: 180 − 65 = 115°.

(ix) A transversal cuts two parallel lines. Each pair of co-interior angles (interior angles on the same side) adds up to:

(a) 180° (b) 90°
(c) 360° (d) a different value each time

> **Ans: (a)** Co-interior angles of parallel lines are supplementary.

(x) The sum of the first 10 odd numbers is:

(a) 55 (b) 110 (c) 90 (d) 100

Ans: (d) Sum of the first n odd numbers is n^2: $10^2 = 100$.

(xi) Which three lengths can be the sides of a triangle?

(a) 4 cm, 5 cm, 9 cm (b) 7 cm, 8 cm, 10 cm
(c) 3 cm, 3 cm, 7 cm (d) 6 cm, 6 cm, 13 cm

Ans: (b) Each sum of two sides must exceed the third: 7 + 8 > 10 holds; in the others the two shorter sides add up to at most the longest.

(xii) The value of 2/3 ÷ 4/9 is:

(a) 3/2 (b) 8/27 (c) 4/3 (d) 1/6

Ans: (a) 2/3 × 9/4 = 18/12 = 3/2.

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Write 5 crore 8 lakh 40 thousand in figures using commas (i) in the Indian system and (ii) in the international system.

(i) 5,08,40,000 (1 mark)
(ii) 50,840,000 (1 mark)

3. Evaluate: 48 ÷ (4 + 2 × 2) + 3 × 2. Show each step.

Brackets first: 4 + 2 × 2 = 4 + 4 = 8 (1 mark)
48 ÷ 8 + 3 × 2 = 6 + 6 = 12 (1 mark)

4. A ribbon 3.5 m long is cut into pieces of 0.25 m each. How many pieces are obtained?

3.5 ÷ 0.25 = 350 ÷ 25 (1 mark) = 14 pieces (1 mark)

OR

Subtract 4.875 from 10.2.

10.200 − 4.875 (aligning decimal points) (1 mark) = 5.325 (1 mark)

5. Simplify: 5x + 3y − 2x + 4y.

Group like terms: (5x − 2x) + (3y + 4y) (1 mark)
= 3x + 7y (1 mark)

6. Two straight lines intersect and one of the four angles formed is 40°. Find the other three angles, giving reasons.

Vertically opposite angle = 40° (1 mark)
Each of the other two forms a linear pair with 40°: 180° − 40° = 140° and 140° (1 mark)

7. Start with 3524 and carry out the Kaprekar routine (arrange digits in descending and ascending order and subtract) until you reach 6174.

5432 − 2345 = 3087 (1 mark)
8730 − 0378 = 8352; 8532 − 2358 = 6174 (1 mark)

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. In triangle ABC the angles are in the ratio ∠A : ∠B : ∠C = 2 : 3 : 4. Find the three angles and state whether the triangle is acute-angled, right-angled or obtuse-angled.

Angle sum 180°: 2k + 3k + 4k = 180 ⇒ k = 20 (1 mark)
∠A = 40°, ∠B = 60°, ∠C = 80° (1 mark)
All angles are less than 90°, so it is acute-angled (1 mark)

OR

The exterior angle of a triangle is 110° and one of its interior opposite angles is 45°. Find the other interior opposite angle and the adjacent interior angle.

> Exterior angle = sum of the two interior opposite angles (1 mark)
> Other opposite angle = 110° − 45° = 65° (1 mark)
> Adjacent interior angle = 180° − 110° = 70° (1 mark)

9. Meena's mother needs 3/4 cup of flour for one batch of rotis. She has 4 1/2 cups of flour. (i) How many batches can she make? (ii) She also uses 2/3 of a 9/10 kg pack of ghee; how many kg is that?

> (i) 9/2 ÷ 3/4 = 9/2 × 4/3 (1 mark) = 6 batches (1 mark)
> (ii) 2/3 × 9/10 = 18/30 = 3/5 kg (1 mark)

10. A city has 12,50,000 houses and every house uses about 480 litres of water a day. Find the total water used in a day (i) in litres, written in the Indian system and in crore, and (ii) in the international system (in millions).

> 1,250,000 × 480 = 600,000,000 (1 mark)
> (i) 60,00,00,000 litres = 60 crore litres (1 mark)
> (ii) 600,000,000 = 600 million litres (1 mark)

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

Questions 11 to 13 carry 5 marks each.

11. Sunita buys 2.5 kg rice at ₹ 48.60 per kg and 1.25 kg sugar at ₹ 44.80 per kg. (i) Find the cost of the rice. (ii) Find the cost of the sugar. (iii) Find the total bill. (iv) She pays with a ₹ 200 note; how much change does she get? (v) How many 50-paise coins make up the cost of the rice?

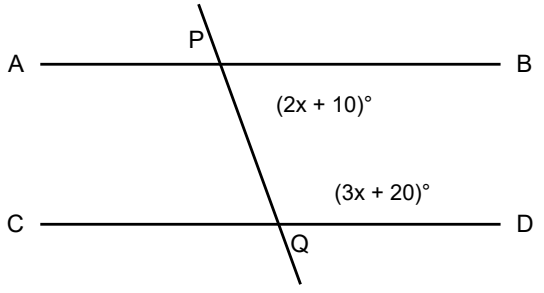
> (i) 2.5 × 48.60 = ₹ 121.50 (1 mark)
> (ii) 1.25 × 44.80 = ₹ 56.00 (1 mark)
> (iii) 121.50 + 56.00 = ₹ 177.50 (1 mark)
> (iv) 200 − 177.50 = ₹ 22.50 (1 mark)
> (v) 121.50 ÷ 0.50 = 243 coins (1 mark)

OR

(i) Write 7/8 as a decimal. (ii) Write 0.375 as a fraction in simplest form. (iii) Arrange 0.6, 0.59, 0.605 and 0.06 in ascending order. (iv) Find 4.56 × 100. (v) Find 7.5 ÷ 10.

> (i) 0.875 (1 mark)
> (ii) 375/1000 = 3/8 (1 mark)
> (iii) 0.06 < 0.59 < 0.6 < 0.605 (1 mark)
> (iv) 456 (1 mark)
> (v) 0.75 (1 mark)

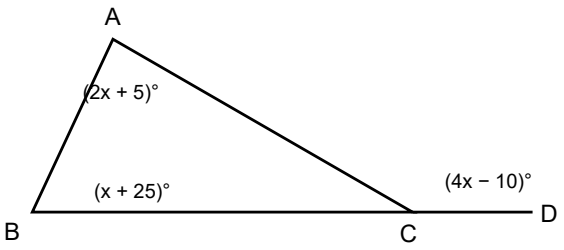
12. In the figure, AB ∥ CD and PQ is a transversal. ∠BPQ = (2x + 10)° and ∠DQP = (3x + 20)°. (i) Name the relation between these two angles. (ii) Find x. (iii) Find ∠BPQ and ∠DQP. (iv) Find ∠APQ.



(i) Co-interior angles (interior angles on the same side), so they are supplementary (1 mark)
(ii) $2x + 10 + 3x + 20 = 180 \Rightarrow 5x = 150 \Rightarrow x = 30$ (2 marks)
(iii) $\angle BPQ = 70°$, $\angle DQP = 110°$ (1 mark)
(iv) $\angle APQ = 180° - 70° = 110°$ (linear pair) (1 mark)

OR

In the figure, side BC of triangle ABC is extended to D. $\angle A = (2x + 5)°$, $\angle B = (x + 25)°$ and the exterior angle $\angle ACD = (4x - 10)°$. (i) State the exterior angle property. (ii) Find x. (iii) Find $\angle A$ and $\angle B$. (iv) Find $\angle ACB$.



(i) An exterior angle equals the sum of the two interior opposite angles (1 mark)
(ii) $4x - 10 = 2x + 5 + x + 25 \Rightarrow x = 40$ (2 marks)
(iii) $\angle A = 85°$, $\angle B = 65°$ (1 mark)
(iv) $\angle ACB = 180° - 150° = 30°$ (check: 85 + 65 + 30 = 180) (1 mark)

13. Squares are made in a row using matchsticks: 1 square needs 4 sticks, 2 squares need 7 sticks, 3 squares need 10 sticks. (i) How many sticks are needed for 5 squares? (ii) Write a rule (expression) for n squares. (iii) How many squares can be made with exactly 100 sticks? (iv) Can exactly 50 sticks make a row of squares with none left over? Why? (v) Is the number of sticks for n squares always odd? Check for n = 1, 2, 3, 4.

(i) $3 \times 5 + 1 = 16$ sticks (1 mark)
(ii) $3n + 1$ (1 mark)
(iii) $3n + 1 = 100 \Rightarrow n = 33$ (1 mark)
(iv) $3n + 1 = 50 \Rightarrow 3n = 49$, not a whole number; so No (1 mark)
(v) 4, 7, 10, 13 are alternately even and odd, so No (1 mark)

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. Rajasthan is the largest state of India by area. According to the 2011 Census, its population was 6,85,48,437 and its area is 3,42,239 square kilometres.

(i) Write the population correct to the nearest lakh.

6,85,48,437 lies between 685 lakh and 686 lakh and 48,437 is less than half a lakh, so ≈ 6,85,00,000 (685 lakh)

(ii) How many crore people is this, approximately (to two decimal places)?

≈ 6.85 crore

(iii) Find the approximate population per square kilometre (nearest whole number).

68,548,437 ÷ 342,239 ≈ 200.3 (1 mark) ≈ 200 persons per km² (1 mark)

OR

(iii) If the population increases by 25,00,000, write the new population in the Indian system.

6,85,48,437 + 25,00,000 (1 mark) = 7,10,48,437 (1 mark)

15. A triangular flower bed ABC in a school garden has AB = 9 m and BC = 7 m. The third side CA = x metres is a whole number. The angles at A and B are 50° and 70°. The gardener extends side BC beyond C to lay a path.

(i) Can CA be 17 m? Give a reason.

> No. 9 + 7 = 16 is less than 17, violating the triangle inequality.

(ii) Write the greatest and the smallest whole number value of x.

> 2 < x < 16, so x can be at most 15 and at least 3.

(iii) Find ∠C and the exterior angle at C.

> ∠C = 180° − (50° + 70°) = 60° (1 mark); exterior angle = 180° − 60° = 120° (= 50° + 70°) (1 mark)

OR

(iii) If a similar flower bed is isosceles with a vertex angle of 40°, find its base angles.

> Base angles are equal: (180° − 40°) ÷ 2 (1 mark) = 70° each (1 mark)

16. Kavya's mother has 12 1/2 litres of milk in a can. She pours it into glasses, each holding 1/4 litre.

(i) How many glasses can she fill?

> 25/2 ÷ 1/4 = 25/2 × 4 = 50 glasses

(ii) Write the reciprocal of 5/8.

> 8/5

(iii) She sells 3/5 of the milk. How many litres are left?

> Left = 2/5 of 25/2 (1 mark) = 5 litres (1 mark)

OR

(iii) Each glass is sold at ₹ 15. Find the money received if 3/5 of the milk is sold.

> Sold = 3/5 × 25/2 = 7.5 L = 30 glasses (1 mark); 30 × 15 = ₹ 450 (1 mark)